Common random number simulation as the optimal coupling for MCMC

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Abstract

In this paper, we demonstrate that common random number simulation (CRN) applied to coordinate-wise monotone iterated random functions (IRF) represents the optimal Wasserstein coupling of an n-step Markov chain (MC). This finding generalizes the results of [16], which establishes sufficient conditions for when CRN serves as the variance reduction coupling between functions of random variables. We provide specific examples of MCs where CRN is optimal, as well as cases where it is not. For those MCs where CRN does not yield the optimal coupling, we illustrate how CRN, applied to equimeasurable transformations, can still result in lower expected distances. Additionally, we argue why Wasserstein distance is the natural metric of choice when studying CRN and discuss the relationship between CRN, backward processes, and geometric convergence.

Contents

1 Introduction **2**

2 Background **5**

3 Applying CRN to quantile functions **7**

4 Conditions when CRN is an optimal coupling **10**

4.1 Conditions when CRN is an optimal coupling on the first iteration of a Markov chain 10

4.2 Conditions when CRN is an optimal coupling on the nth iteration of a Markov chain 13

4.3 Examples 15

5 When monotone conditions are not satisfied **21**

5.1 Equimeasurable functions 22

5.2 Example 23

6 Connection between CRN and the backwards process **27**

6.1 Connection between CRN and the contraction condition 29

7 Appendix **30**

1 Introduction

Markov chain Monte Carlo (MCMC) is an integral part of statistical computing and Bayesian statistics is its most notable beneficiary [18, 5]. An important question when simulating MCMC is how quickly does the chain converge to the stationary distribution of interest [18]. The Wasserstein distance, a measurement of distributions, is a popular method for evaluating whether a chain is close to stationarity [28, 15]. It scales well in high dimensions [28] and metrizes convergence in distribution (see Theorem 6.9 of [39]).

Various methods for generating theoretical upper bounds on the Wasserstein distance between the distribution of a Markov chain and its stationary distribution exist [28, 27]. However theoretical bounds involve complex calculation, are difficult to implement, and do

not exist for many MCMC models of interest [21, 27, 35, 37].

An alternative method for estimating upper bounds on the Wasserstein distance is through common random number (CRN) simulation [16, 19, 36] alternatively called synchronous coupling [1, 6, 19]. CRN simulation translates to using the same random transform to generate two copies of a Markov chain (see Section 2 for more details) and is straightforward to implement.

CRN simulation was proposed as the variance reduction coupling in as early as 1934 [17] and has been applied over a wide range of models including MCMC [36, 3, 22]. In certain examples the contraction condition, which uses CRN (see Section 6 for more details between the connection of CRN and the contraction condition), generates either the smallest geometric convergence rate [26, 35, 29] or appears to be more tighter than drift and minorization bounds [36, 20].

CRN has been shown to be the optimal coupling for a variety of stochastic process examples. Namely, in [6], it is shown that under certain conditions two reflected Brownian motion processes simulated using CRN converge to 0 almost surely. In [1] it is shown that CRN simulation is an optimal Wasserstein coupling between the solutions of SDEs. In [10] it is shown that CRN simulation is the maximal total variation coupling on the independent Metropolis algorithm.

The optimality for CRN applied to MCMC was first discussed in [16]. In $\mathbb{R}^d, d \geqslant 1$, Proposition 2.2 of [16] states that CRN applied to two coordinate-wise increasing transforms of two random variables minimizes the Euclidean distance. In this paper, we provide sufficient conditions for when CRN is an optimal Wasserstein coupling on the nth iteration of a Markov chain in $\mathbb{R}^d$ defined over general distances (see Theorem 4.5). We generalize the results of [16] to quasi-antitone distance functions, show how CRN optimality can be applied to Markov chains, and add more details to the proof.

We provide examples where CRN is the optimal coupling for a 1-step chain and an n-step chain. We show that CRN could be the optimal coupling for a 1-step chain, but not

an n-step chain (see Example 4.1 compared to Numerical Example 5.1). We also show that even though CRN is the optimal coupling for a 1-step chain, it does not necessarily imply that two copies simulated using CRN will converge in expectation (see Example 4.1).

In section 3, we start off by summarizing the useful properties of CRN when the random transform is the quantile function (aka the inverse distribution function). We justify why the Wasserstein distance is the natural measure for studying the convergence properties of random values simulated using CRN. In particular, for a Markov chain $\{X_n\}_{n\geqslant 1}$ defined in the metric space $(\mathcal{X}, k)$, if we can write $X_n = F_n^{-1}(U_n)$ where F_n is the distribution function of X_n and $U_n \sim Unif(0,1)$, then applying CRN to two copies of a Markov chain means that convergence in distribution in $P_p(\mathcal{X})$ is equivalent to convergence in the k mean when simulated using CRN (Proposition 3.2). Further, convergence in distribution is equivalent to almost sure convergece of two chains simulated using CRN (Proposition 3.3).

In practice, we do not know the quantile function of X_n. Rather we define the Markov chain as a sequence of iterated random functions (IRF). That is, we can write $X_n = f(\theta_n, X_{n-1})$ where θ_n is a random variable. In Section 4.1, we identify when CRN is an optimal coupling on the first iteration of a Markov chain. In Theorem 4.4 we show that if $f(\vec{\theta}_n, \vec{X}_{n-1})$ is increasing coordinate-wise for each $\theta_{i,n} \in \vec{\theta}_n$ for any $\vec{X}_{n-1}$ then CRN on the first iteration of a Markov chain is an optimal coupling.

In Section 4.2 we extend the results of Section 4.1 to provide sufficient conditions for when CRN is the optimal coupling on the nth iteration of a Markov chain. In Theorem 4.5 we show that if $f(\vec{\theta}, \vec{x})$ is increasing coordinate-wise with respect to $\vec{\theta}$ and $\vec{x}$, then CRN on the nth iteration of a Markov chain is an optimal coupling. This means that if two copies of a Markov chain, $\{\vec{X}_n\}_{n\geqslant 1}$ and $\{\vec{Y}_n\}_{n\geqslant 1}$, converge in distribution to a stationary distribution π, then $k(\vec{X}_n, \vec{Y}_n)$ converges in probability to 0 when generated by CRN (see Corollary 1). If two copies of a Markov chain, $\{\vec{X}_n\}_{n\geqslant 1}$ and $\{\vec{Y}_n\}_{n\geqslant 1}$, are geometrically ergodic, then $k(X_n, Y_n)$ converges almost surely to 0 when generated by CRN (see Corollary 2).

In Section 6, we establish the connection between CRN and backwards processes by

showing in Theorem 6.1 that the expected distance between the two coupled chains is equal to the expected distance between the backwards process at iterations n and $n+1$. We also cover how contraction rates implicitly assume CRN in the joint coupling.

First we introduce notation in the Section 2. This section should be referenced if the reader wants clarity on notation throughout the paper.

2 Background

Markov chain notation: Let $\{X_n\}_{n\geqslant 1}$ be a Markov chain such that X_n can be written as an iterative random function ie. $X_n = f(\vec{\theta}_n, X_{n-1}) \in \mathbb{R}$ where $\vec{\theta}_n \in \mathbb{R}^d, n \geqslant 1$ are in the probability space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mu)$. Note that $\vec{\theta}_n$ are i.i.d. w.r.t n and independent w.r.t. d. That is, for i.i.d. random variables $(\vec{\theta}_1, \ldots, \vec{\theta}_n) = \Theta_n$ on the support μ and initial value x_0

$$X_n = f(\vec{\theta}_n, f(\vec{\theta}_{n-1}, \ldots, f(\vec{\theta}_1, x_0)\ldots)) = f_n(\Theta_n, x_0) \tag{1}$$

Any time-homogeneous Markov chain can be represented as an IRF [38]. X_n is the value of the Markov chain at iteration n.

$\mathcal{L}(X_n)$ denotes the law of X_n and $\mathcal{L}(X_n) \to \pi$ denotes weak convergence of the law of X_n to a stationary distribution π.

The Markov chain $\{\vec{X}_n\}_{n\geqslant 1} \in \mathbb{R}^q, q \geqslant 1$ is a vector form of the initial chain where for each vector coordinate $i \in \{1, \ldots, q\}$ we have $X_n^{(i)} = f^{(i)}(\vec{\theta}_n, \vec{X}_{n-1})$.

CRN CRN simulation occurs when we simulate two Markov chains $\{X_n\}_{n\geqslant 1}$ and $\{Y_n\}_{n\geqslant 1}$ with different initial values $x_0 \neq y_0$, but with the same transition functions. That is, $\vec{\theta}_n$ are the same in $X_n = f(\vec{\theta}_n, X_{n-1})$ and $Y_n = f(\vec{\theta}_n, Y_{n-1})$, $n \geqslant 1$. In different notation, Θ_n is the same in $X_n = f_n(\Theta_n, x_0)$ and $Y_n = f_n(\Theta_n, y_0)$, $n \geqslant 1$. CRN is unique to the IRF, but not the the Markov chain [16]. Some texts limit CRN to the inverse transform of the distribution function [16, 19] like what is done in Section 3.

Distance function k: Let k be a distance function.

k is quasi-antitone (in other words $-k$ is supermodular) if for all $x \leqslant x'$, $y \leqslant y'$ π-a.e.,

$$k(x,y) + k(x',y') - k(x,y') - k(x',y) \leqslant 0$$

We will need k to be quasi-antitone to prove that CRN is an optimal coupling. We write that k is QAC if k is a quasi-antitone continuous distance function.

Theorem 3 of [9] tells us that functions $f(x-y)$ where f is convex and $f(0) = 0$ are QAC. The ℓ^p norm is defined as follows, $||\vec{x}||_p = (\sum_{i=1}^{q} x_i^p)^{1/p}$ and is a QAC distance function.

Wasserstein distance We define $J(X,Y)$ as a joint distribution function of X, Y and $\mathcal{J}(\mu,\nu) = \{J(X,Y) \mid \mathcal{L}(X) = \mu, \mathcal{L}(Y) = \nu\}$ to be the set of all joint distribution functions with marginals μ, ν. The Wasserstein distance of order p is $W_{k,p}(\mu,\nu) = \inf_{J \in \mathcal{J}(\mu,\nu)} E_J[k(X,Y)^p]^{1/p}$ for $p \in [1,\infty)$.

$J \in \mathcal{J}(\mu,\nu)$ is an optimal coupling if $E_J[k(X,Y)^p]^{1/p} = W_p(\mu,\nu)$ [39]. That is J is the joint distribution such that the expected distance between X and Y using joint distribution J is equal to the Wasserstein distance.

We define the Wasserstein space of order p, $P_p(\mathcal{X})$, as the space of Borel probability measures on the metric space $(\mathcal{X}, k)$ such that for any $\mu \in P_p(\mathcal{X})$, $\int_{\mathcal{X}} k(x_0,x)^p \mu dx < \infty$ for some $x_0 \in \mathcal{X}$.

Define a sequence of random variables $\{X_n\}_{n \geqslant 1}$ in a Polish metric space $(\mathcal{X}, k)$ with distributions in $\mathcal{L}(X_n) \in P_p(\mathcal{X})$. Define $\mathcal{L}(X) = \pi \in P_p(\mathcal{X})$. X_n converges weakly to π in $P_p(\mathcal{X})$ (denoted as $\mathcal{L}(X_n) \to \pi$ in $P_p(\mathcal{X})$) if (1) $\mathcal{L}(X_n) \to \pi$ and (2) for some $x_0 \in \mathbb{R}$, $E[k(x_0,X_n)^p] \to E[k(x_0,X)^p]$ (see alternative conditions in Definition 6.8 of [39]).

3 Applying CRN to quantile functions

We first discuss the simplest case of CRN: when random variables $X, Y \in \mathbb{R}$ are written as the quantile transform of a uniform random variable. We define the quantile function as $F^{-1}(u) = \inf\{x \mid u \leqslant F(x)\}$, and our random variables as functions of the quantile function, $X = F_x^{-1}(U) \sim \mu$ and $Y = F_y^{-1}(V) \sim \nu$ where $U, V \sim Unif(0,1)$). We can specify the interval of expected values for distance $E[k(X,Y)]$ when $J \in \mathcal{J}(\mu,\nu)$ in Proposition 3.1.

Proposition 3.1 lays the foundation for the rest of the theorems in this paper. Namely Proposition 4.1 uses Proposition 3.1 to prove that CRN is the optimal coupling when X is an increasing function of U.

Proposition 3.1 is important for generalizing optimality properties of random variables simulated using CRN, because it specifies the set of distance functions k where CRN is the optimal coupling.

Proposition 3.1 (Theorem 2 of [7])**.** *Suppose that $X, Y \in \mathbb{R}$ are two random variables with corresponding distribution functions F_x, F_y such that $X \sim \mu$ and $Y \sim \nu$. Suppose that k is QAC and that $\mu, \nu \in P_p(\mathbb{R})$. Then, the set of expected distances of $k(X,Y)$ with joint distribution $J \in \mathcal{J}(\mu,\nu)$ is equal to the following interval,*

$$\left\{E_J[k(X,Y)] \mid J \in \mathcal{J}(\mu,\nu)\right\} = \left[E[k(F_x^{-1}(U), F_y^{-1}(U))], E[k(F_x^{-1}(U), F_y^{-1}(1-U))]\right]$$

Proof. Note that condition (a) of Theorem 2 of [7] is satisfied, because k is symmetric and $\mu, \nu \in P_p(\mathbb{R})$ and so for some $x_0 \in \mathbb{R}$, $E[k(X,Y)] \leqslant E[k(X,x_0)] + E[k(x_0,Y)] < \infty$. □

Proposition 3.2 summarizes why we focus on Wasserstein distance to study convergence of CRN applied to quantile functions. Namely, the Wasserstein distance metrizes weak convergence in $P_p(\mathbb{R})$. This along with the fact that in one dimension the Wasserstein distance admits the quantile representation, $W_p(\mu,\nu) = E[k(F_\mu^{-1}(U), F_\nu^{-1}(U))^p]^{1/p}$ when k is QAC means that CRN applied to the quantile function metrizes convergence in distribution in $P_p(\mathbb{R})$.

The importance of this observation lies in the fact that it identifies the Wasserstein distance as a natural metric for studying convergence of random variables generated via CRN. This is in contrast to the total variation distance, which is the metric used to analyze CRN convergence in [22, 10]. Total variation distance does not metrize convergence in distribution. Indeed, convergence in total variation distance requires that $P(X = Y) \to 1$, which is equivalent to the existence of a coupling under which the two random variables become equal with probability tending to one. In most applications of CRN for real-valued random variables, however, the coupled variables are not equal almost surely; rather, they are constructed to be strongly positively dependent.

Proposition 3.2. *Define the random variables $\{X_n\}_{n\geqslant 1}$ and X in a Polish metric space $(\mathbb{R}, k)$. The following are equivalent where $U \sim \text{Unif}(0,1)$.*

1. $\mathcal{L}(X_n) \to \pi$ *in* $P_p(\mathbb{R})$
2. $W_p(\mathcal{L}(X_n), \mathcal{L}(X)) \to 0$
3. $E\big[k(F_{X_n}^{-1}(U), F_X^{-1}(U))^p\big]^{1/p} \to 0$

Proof. (1) $\iff$ (2): by Theorem 6.9 of [39].

(2) $\iff$ (3): By Proposition 3.1, the Wasserstein distance is attained when CRN is applied for each n,

$$\begin{aligned} W_p(\mathcal{L}(X_n), \mathcal{L}(X)) &= \left(\inf_{J \in \mathcal{J}(\mathcal{L}(X_n), \mathcal{L}(X))} E_J\big[k(X_n, X)^p\big] \right)^{1/p} \\ &= E\big[k(F_{X_n}^{-1}(U), F_X^{-1}(U))^p\big]^{1/p} \end{aligned}$$

□

Proposition 3.3 shows that CRN applied to quantile functions can be in fact stronger than convergence in Wasserstein distance. Namely almost sure convergence of CRN applied to quantile functions metrizes convergence in distribution (as opposed to convergence in distribution in $P_p(\mathbb{R})$).

Proposition 3.3 can be interpreted as a corollary of Skorohod's theorem.

Proposition 3.3. *Define the random variables $\{X_n\}_{n\geqslant 1}$ and stationary distribution π in the metric space $(\mathbb{R}, |\cdot|)$. Then the following holds,*

$$\mathcal{L}(X_n) \to \pi \iff |F_n^{-1}(U) - F^{-1}(U)| \to 0 \text{ w.p. 1}$$

Where F_n and F are the CDF of $\mathcal{L}(X_n)$ and π, respectively.

Proof. See proof in Subsection 7.1, which closely follows the proof of Skorohod's theorem (see Theorem 25.6 in [2]). □

The following example provides a case where $\mathcal{L}(X_n) \to \pi$ in distribution but not in $P_p(\mathbb{R})$, so Theorem 3.2 does not hold, but Theorem 3.3 holds.

Example 3.1. Let $\{X_n\}_{n\geqslant 1}$ be a sequence such that for each $n \geqslant 1$, $P(X_n = n) = \frac{1}{n}$ and $P(X_n = 0) = 1 - \frac{1}{n}$. $\mathcal{L}(X_n)$ converges weakly to the Dirac mass at point 0, $\mathcal{L}(X_n) \to \delta_0$. In this case F_n and F_n^{-1} are defined as follows,

$$F_n(x) = \begin{cases} 0 & x < 0 \\ 1 - \frac{1}{n} & 0 \leqslant x < n \\ 1 & n \leqslant x \end{cases} \qquad F_n^{-1}(u) = \begin{cases} 0 & 0 \leqslant u \leqslant 1 - \frac{1}{n} \\ n & 1 - \frac{1}{n} < u \leqslant 1 \end{cases}$$

$\mathcal{L}(X_n)$ does not converge weakly to δ_0 in $P_1(\mathbb{R})$: Fix a point x_0, $\lim_{n\to\infty} E[|x_0 - X_n|] = \lim_{n\to\infty} |x_0 - 0|(1 - \frac{1}{n}) + |x_0 - n|\frac{1}{n} = |x_0| + 1 \neq |x_0 - 0|$.

Proposition 3.3 applies: First note that $F_{\delta_0}^{-1}(u) = 0$ for all $u \in (0, 1]$. Fix $u \in (0, 1)$, then for $n > \frac{1}{1-u}$, $F_n^{-1}(u) = 0 = F_{\delta_0}^{-1}(u)$. So F_n^{-1} converges pointwise to $F_{\delta_0}^{-1}$ for all $u \in (0, 1)$. Since $P(U \in (0, 1)) = 1$, $|F_n^{-1}(U) - F_{\delta_0}^{-1}(U)| \to 0$ w.p. 1.

Bullet (3) of Proposition 3.2 does not apply: $E[|X_n - 0|] = |n-0|\frac{1}{n} + |0-0|(1-\frac{1}{n}) = 1 \not\to 0$.

While the results related to the convergence of quantile transforms are elegant, they are generally impractical for studying the optimality of CRN applied to Markov chains. We usually do not know the quantile function of X_n. Further, we can only apply quantile

transforms when $X \in \mathbb{R}$ (see Section 2 of [32]). In the next section we will provide sufficient conditions for when CRN is the optimal coupling between two IRFs in $\mathbb{R}^q$.

4 Conditions when CRN is an optimal coupling

4.1 Conditions when CRN is an optimal coupling on the first iteration of a Markov chain

The following theorem provides conditions under which the expected distance using CRN between two non-decreasing random functions in $\mathbb{R}$ is equal to the Wasserstein distance. This theorem is a generalization of Proposition 2.2 of [16], which says that the covariance between two non-decreasing random functions is maximized when generated by CRN. Before we specify the theorem we introduce the concept of monotonicity with respect to θ.

Definition 4.1 (Monotone in θ)**.** Suppose that $f(\theta, x)$ is an IRF where $\theta \sim \mu$ and $x \in \mathbb{R}$. The function f is considered to be monotone in θ, if for all $\theta_1 \leqslant \theta_2$ μ-a.e. and for all x π-a.e.

$$f(\theta_1, x) \leqslant f(\theta_2, x) \tag{2}$$

Proposition 4.1 (Conditions when CRN is an optimal coupling, $\theta \in \mathbb{R}$ and $X \in \mathbb{R}$)**.** *Let $X = f(\theta_1, x)$ and $Y = f(\theta_2, y)$ be defined on the Polish metric space $(\mathbb{R}, k)$ where k is QAC, $\mathcal{L}(X), \mathcal{L}(Y) \in P_p(\mathbb{R})$, and θ_1, θ_2 are identically distributed random variables. Suppose that f is monotone and right-continuous in θ. Then CRN is the optimal coupling,*

$$W_p(\mathcal{L}(X), \mathcal{L}(Y)) = E\big[k(f(\theta, x), f(\theta, y))^p\big]^{1/p} \tag{3}$$

The proof is in Section 7.2.

Proposition 4.1 states that if the functions $f(\cdot, x)$ and $f(\cdot, y)$ are increasing and defined on the real line, then the expected distance using CRN equals the Wasserstein distance.

Connection to Brenier's Theorem Within the proof of Proposition 4.1, we used Proposition 3.1, however, we could have also used Brenier's theorem, with specified $k(x, y) = (x - y)^2$, which coincides with Proposition 2.2 of [16]. The following is a simplified statement of Brenier's theorem in $\mathbb{R}$.

Proposition 4.2 (Brenier's Theorem (Theorem 2.1 of [25], [4]))**.** *Suppose $X, Y \in \mathbb{R}$ are random variables, $k(x, y)^2 = (x - y)^2$, and $\mathcal{L}(X), \mathcal{L}(Y) \in P_p(\mathbb{R})$. If X has a density with respect to the Lebesgue measure, then the random variables that generate the optimal joint distribution $J^*(\mathcal{L}(X), \mathcal{L}(Y))$ can be defined as X and $Y = T(X)$ where T is the derivative of a convex function.*

Remark 4.1. Brenier's theorem implies that if g, h are increasing functions, then CRN is the optimal coupling.

First note that $k(x, y)^2 = (x - y)^2$ is a QAC function. Next note that if g is an increasing function, then $g^{-1}(x) = \inf\{u \mid g(u) \geqslant x\}$ is an increasing function. Let $X = g(U)$ and $Y = h(V)$ where $U, V \sim \text{Unif}(0, 1)$. The coupling $J^*(\mathcal{L}(X), \mathcal{L}(Y))$ where $Y = h(g^{-1}(X))$ is optimal according to Brenier's theorem, because $T = h \cdot g^{-1}$ is an increasing function and thus is the derivative of a convex function. Now we remark that $g^{-1}(X) \sim \text{Unif}(0, 1)$ and so for $U \sim \text{Unif}(0, 1)$, $X = g(U)$ and $Y = h(U)$ thus implying that CRN is the optimal coupling. This is only a sketch, however, a more thorough explanation can be found in Section 2.1 of [32] and Section 2.3.1 of [19].

If our Markov chain is in $\mathbb{R}^q, q \geqslant 1$, the state space is no longer naturally endowed with a total order and so the definition of monotonicity in $\mathbb{R}$, Definition 4.3, needs additional conditions. Brenier specifies the optimal mapping to be the gradient of a convex function, which can be applied to Markov chains in $\mathbb{R}^q$. In [16] the optimal coupling occurs when CRN is applied to functions that are coordinate-wise increasing.

Markov chains that have coordinate-wise increasing transition functions are related to stochastic and realizable monotonicity [13, 14] and stochastic dominance [23], which have

themselves been shown to be related to optimal coupling. In [13], the inverse probability transform realizes the monotonicity of a stochastically monotone system, which can then be applied to perfect sampling algorithms. In [23] (see their Section 2), it is shown that the total variation distance maximal coupling is a subset of the joint distributions that realize stochastic dominance. This paper, similar to [16], focuses on the property of coordinate-wise increasing functions to determine optimal coupling.

Next we will provide sufficient conditions for when CRN is the optimal coupling on the first iteration of a Markov chain when $\vec{\theta} \in \mathbb{R}^d$, $d \geqslant 1$. This theorem closely resembles Proposition 2.2 of [16], but adds more details to the proof, applies the concepts to IRFs and generalizes the distance functions from the ℓ^2 norm to QAC distances.

Definition 4.2 (Monotone in θ_i)**.** Let $f(\vec{\theta}, \vec{x}) \in \mathbb{R}^q$ be an iterative random function where $\vec{x} \in \mathbb{R}^q$, $\vec{\theta} \in \mathbb{R}^d$, and $\theta \sim \mu$ ($\theta_i \sim \mu_i$). f is considered to be monotone in θ_i, if for all $\theta_i \leqslant \theta_i'$ μ_i-a.e. and $\vec{\theta}_{-i}$ μ_{-i}-a.e., $\vec{x}$ π-a.e.

$$f^{(l)}((\theta_1, \ldots, \theta_i, \ldots, \theta_d), \vec{x}) \leqslant f^{(l)}((\theta_1, \ldots, \theta_i', \ldots, \theta_d), \vec{x}) \tag{4}$$

for all $l \in \{1 \ldots q\}$

Definition 4.3 (Coordinate-wise monotone in $\vec{\theta}$)**.** The function f is considered to be monotone in $\vec{\theta}$, if it is monotone in θ_i for all $i \in \{1, \ldots, d\}$.

Proposition 4.3 (Conditions when CRN is an optimal coupling, $\vec{\theta} \in \mathbb{R}^d, d \geqslant 1$ and $X \in \mathbb{R}$. Extension of Proposition 2.2 of [16].)**.** *Let X and Y be two copies of a random variable defined on a metric space $(\mathbb{R}, k)$ where k is QAC and $\mathcal{L}(X), \mathcal{L}(X) \in P_p(\mathbb{R})$. Suppose that $X = f(\vec{\theta}, x)$ ($Y = f(\vec{\theta}', y)$) can be written as a random function where $\vec{\theta} \in \mathbb{R}^d$ is a random vector $\vec{\theta} \sim \mu$ ($\vec{\theta}' \sim \mu$) with independent coordinates $\theta_i \perp\!\!\!\perp \theta_j$ for $i \neq j$. Suppose further that f is coordinate-wise monotone in $\vec{\theta}$. Then the Wasserstein distance over the unconditional*

X and Y is attained when CRN is applied,

$$W_p(\mathcal{L}(X), \mathcal{L}(Y)) = E[k(f(\vec{\theta}, x), f(\vec{\theta}, y))] \tag{5}$$

The proof is in Section 7.3.

Finally, we generalize Proposition 4.3 to apply to Markov chains that are defined in $\mathbb{R}^q, q \geqslant 1$. Namely, if for each coordinate of $\vec{X}, \vec{Y}$ the infimum distance is attained by using CRN, then the infimum distance over all $\vec{X}, \vec{Y}$ is attained when using CRN.

Theorem 4.4 (Conditions when CRN is an optimal coupling, $k = ||\cdot||_p$, $\vec{\theta} \in \mathbb{R}^d, d \geqslant 1$ and $\vec{X} \in \mathbb{R}^q, q \geqslant 1$.)**.** *Let $\vec{X}$ and $\vec{Y}$ be two copies of a random variable defined on a metric space $(\mathbb{R}^q, ||\cdot||_p), q, p \geqslant 1$ such that $\mathcal{L}(\vec{X}), \mathcal{L}(\vec{Y}) \in P_{||\cdot||_p}(\mathbb{R}^q)$. Suppose that $\vec{X} = f(\vec{\theta}, \vec{x})$ can be written as a random function where $\vec{\theta} \in \mathbb{R}^d$ is a random vector with independent coordinates $\theta_i \perp\!\!\!\perp \theta_j$ for $i \neq j$ and each coordinate $X^{(i)} = f^{(i)}(\vec{\theta}, \vec{x})$. Note that $\vec{Y} = f(\vec{\theta'}, \vec{y})$ where $\mathcal{L}(\vec{\theta}) = \mathcal{L}(\vec{\theta'}) = \mu$.*

Suppose further that $\vec{X}$ and $\vec{Y}$ are coordinate-wise monotone in $\vec{\theta}$. Then the Wasserstein distance of order p is attained when CRN is applied,

$$W_{||\cdot||_p,p}(\mathcal{L}(\vec{X}), \mathcal{L}(\vec{Y})) = E[||f(\vec{\theta}, \vec{x}) - f(\vec{\theta}, \vec{y})||_p^p]^{1/p} \tag{6}$$

The proof is in Section 7.4

4.2 Conditions when CRN is an optimal coupling on the nth iteration of a Markov chain

So far we've provided conditions for when CRN is the optimal coupling on the first iteration of the Markov chain. Next, in Theorem 4.5, we show that if a Markov chain is coordinate-wise monotone with respect to $\vec{\theta}$ and $\vec{x}$ then CRN is an optimal coupling at any iteration of the Markov chain.

Theorem 4.5. *Let $\vec{X}_n, \vec{Y}_n \in \mathbb{R}^q, q \geqslant 1$ be two copies of a Markov chain such that $\vec{X}_n = f(\vec{\theta_n}, \vec{X}_{n-1}) = f_n(\Theta_n, \vec{x}_0)$ and $\vec{Y}_n = f(\vec{\theta'_n}, \vec{Y}_{n-1}) = f_n(\Theta'_n, \vec{y}_0)$ where $\Theta_n = \{\vec{\theta}_1, \vec{\theta}_2, \ldots, \vec{\theta}_n\}$, $\theta_{i,j} \perp\!\!\!\perp \theta_{m,l}$ for any $i, m \in \{1 \ldots n\}$ and $j, l \in \{1, \ldots d\}$.*

Suppose further that $f(\vec{\theta}, \vec{x})$ is monotone in $\vec{\theta}$ and $\vec{x}$, that $\mathcal{L}(\vec{X}_n), \mathcal{L}(\vec{Y}_n) \in P_p(\mathbb{R}^q)$ that k is QAC if $x \in \mathbb{R}$ or $k = ||\cdot||_p$ if $\vec{x} \in \mathbb{R}^q$. Then the Wasserstein distance at iteration n is attained when using CRN,

$$W_k(\mathcal{L}(\vec{X}_n), \mathcal{L}(\vec{Y}_n)) = E[k(f_n(\Theta_n, \vec{x}_0), f_n(\Theta_n, \vec{y}_0))] \tag{7}$$

The proof is in Section 7.5

If we can show that the CRN is an optimal coupling of a Markov process at any iteration, then it follows that if the Markov chains converges to some stationary distribution π, the chains simulated using CRN will converge in probability (Corollary 1). If the process is geometrically ergodic, then two chains simulated using CRN will converge almost surely (Corollary 2).

Corollary 1. *Let $\vec{X}_n, \vec{Y}_n \in \mathbb{R}^q, q \geqslant 1$ be two copies of a Markov chain. If the conditions of Theorem 4.5 are satisfied and $\mathcal{L}(\vec{X}_n), \mathcal{L}(\vec{Y}_n)$ converge to π in $P_p(\mathbb{R}^q)$, then $\{\vec{X}_n\}_{n\geqslant 1}, \{\vec{Y}_n\}_{n\geqslant 1}$ generated using CRN will converge in probability. That is, for $\epsilon > 0$*

$$\lim_{n\to\infty} P(k(f_n(\Theta_n, \vec{x}), f_n(\Theta_n, \vec{y})) \geqslant \epsilon) = 0$$

The proof is in Section 7.6.

Corollary 2. *Let $\vec{X}_n, \vec{Y}_n \in \mathbb{R}^q, q \geqslant 1$ be two copies of a Markov chain. If the conditions of Theorem 4.5 are satisfied and $\vec{X}_n$ is geometrically ergodic, then $\{\vec{X}_n\}_{n\geqslant 1}, \{\vec{Y}_n\}_{n\geqslant 1}$ generated using CRN will converge almost surely.*

$$P(\lim_{n\to\infty} k(f_n(\Theta_n, \vec{x}), f_n(\Theta_n, \vec{y})) = 0) = 1 \qquad a.s.$$

The proof is in Section 7.7.

4.3 Examples

4.3.1 Random logistic maps

Example 4.1 (Random logistic maps)**.** Let $\{X_n\}_{n\geqslant 1} \in [0,1]$ be a random logistic map ([37, 24]). That is for i.i.d. $\beta_n \sim Beta(\alpha+1/2, \alpha-1/2)$ $(\alpha > 1/2)$,

$$X_n = f(\beta_n, x_{n-1}) = 4\beta_n X_{n-1}(1-X_{n-1})$$

For all $\alpha > 1/2$, the random logistic map converges in distribution to the stationary distribution, $Beta(\alpha,\alpha)$, and for $\alpha > 2$, the system is locally contractive (Proposition 7 of [37]) and geometrically ergodic (Theorem 1 of [37]). In Lemma 4.6, we show that CRN is an optimal coupling on the first iteration of the random logistic map (Lemma 4.6), but is not the optimal coupling on a 2-step random logistic map (see Section 5.2).

Figure 1 provides the simulated expected distance when CRN is applied. For all of the α in the figure, the chains converge to the stationary distribution. However, for the red lines, which are proven to be geometrically ergodic, the simulated expected distance converges much faster than the black lines and appear to converge to 0. On the other hand, when $\alpha = 0.6$, the simulated expected distance does not appear to converge to 0 (at iteration 100000, $E[|X_{100k} - X'_{100k}|] \approx 0.343$, which is nearly unchanged from iteration 100, $E[|X_{100} - X'_{100}|] \approx 0.331$). This indicates that CRN being the optimal coupling at one step and converging in distribution are not sufficient conditions for the expected distance to converge to 0 when simulated using CRN.

Lemma 4.6. *CRN is an optimal coupling on the first iteration of the random logistic map when k is QAC.*

Proof. Since $4X_{n-1}(1-X_{n-1}) > 0$ w.p. 1, the function $4\beta_n X_{n-1}(1-X_{n-1})$ is increasing

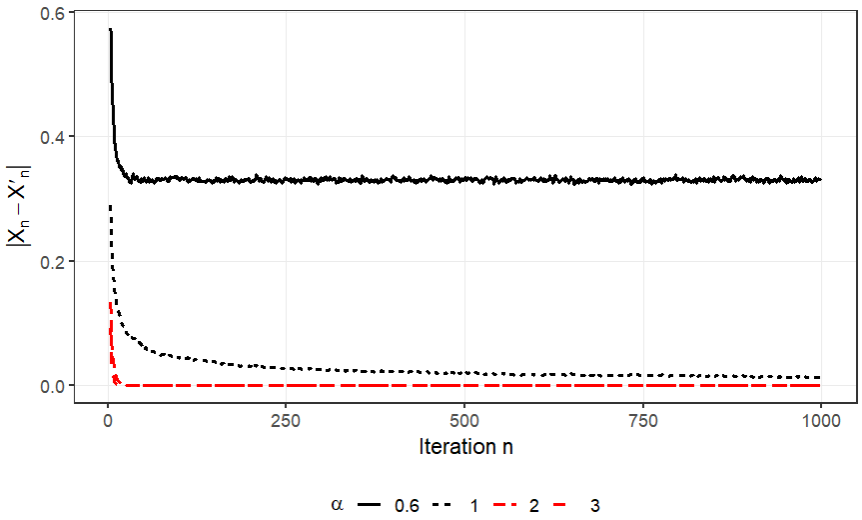


Figure 1: The simulated expected value of $|X_n - X'_n|$ where X_n, X'_n are random logistic maps and CRN is applied ($X_0 = 0.5, X'_0 = 0.125$). For all $\alpha > 1/2$, the chain converges in distribution. For $\alpha \geqslant 2$, the chain is geometrically ergodic. The CRN estimate uses the mean of 10000 simulations.

with respect to β_n. By Proposition 4.1, CRN is an optimal coupling on the first iteration of the Markov chain when k is QAC. □

4.3.2 Zeno's walk

Example 4.2 (Zeno's walk)**.** Let $\{X_n\}_{n\geqslant 1} \in \mathbb{R}$ be a Zeno's walk ([37]). That is for i.i.d. $B_n \sim \text{Bernoulli}(r)$, $X_n = B_n f^+(X_{n-1}) + (1 - B_n) f^-(X_{n-1})$ where,

$$f^+(x) = \begin{cases} x+1 & 0 \leqslant x \\ \frac{x}{2}+1 & -2 \leqslant x \leqslant 0 \\ x+2 & x \leqslant -2 \end{cases} \qquad f^-(x) = \begin{cases} x-1 & x \leqslant 0 \\ \frac{x}{2}-1 & 0 \leqslant x \leqslant 2 \\ x-2 & 2 \leqslant x \end{cases}$$

See Figure 2 for a graphical representation of $f^+(x)$ and $f^-(x)$.

Zeno's walk is locally contractive and geometrically ergodic if $\frac{1}{3} < r < \frac{2}{3}$ [37]. In Lemma 4.7, we show that CRN is the optimal coupling on the nth iteration of Zeno's walk.

Lemma 4.7. *CRN is an optimal coupling on the nth iteration of Zeno's walk when k is QAC.*

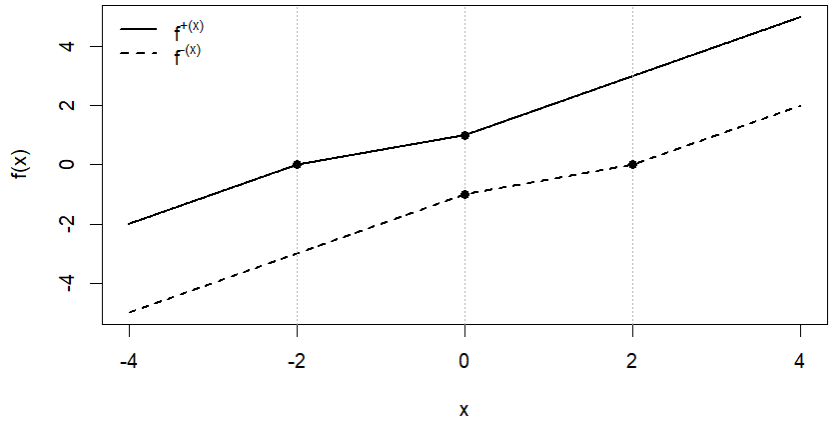


Figure 2: Graph of $f^+(x)$ and $f^-(x)$. It is clear from the graph that $f^+(x) - f^-(x) \geqslant 0$ for any $x \in \mathbb{R}$.

Proof. First we show that CRN is an optimal coupling on the first iteration.

We can re-write $X_n = f^-(X_{n-1}) + B_n(f^+(X_{n-1}) - f^-(X_{n-1}))$. Note that $f^+(x) - f^-(x) > 0$ for all $x \in \mathbb{R}$ (this is clear from Figure 2).This means that for any $X_{n-1} \in \mathbb{R}$, the function $X_n = f^-(X_{n-1}) + B_n(f^+(X_{n-1}) - f^-(X_{n-1}))$ is an increasing function of B_n and so by Proposition 4.1, for k QAC, CRN is an optimal coupling on the first iteration of Zeno's walk.

Next, since $f^+(x), f^-(x)$ are increasing functions of x and $B_n, 1 - B_n \geqslant 0$, the function $B_n f^+(x) + (1 - B_n)f^-(x)$ is an increasing function of x and so, by Theorem 4.5 CRN is the optimal coupling for the nth iteration of the Markov chain. □

Figure 3 displays the expected distance using CRN between two copies of the Zeno walk, X_n and X'_n, initialized at $X_0 = 0$ and $X'_0 = 100$, respectively. By Lemma 4.7, this is the unbiased estimate of the Wasserstein distance.

Moreover, Section 7.2 of [37] asserts that the process fails to converge whenever $r \leqslant 1/3$. This theoretical result is reflected in Figure 3: the black curves (corresponding to $r = 0.1$ and $r = 1/3$) are non-convergent, whereas the red curve (corresponding to $r = 0.5$) converges.

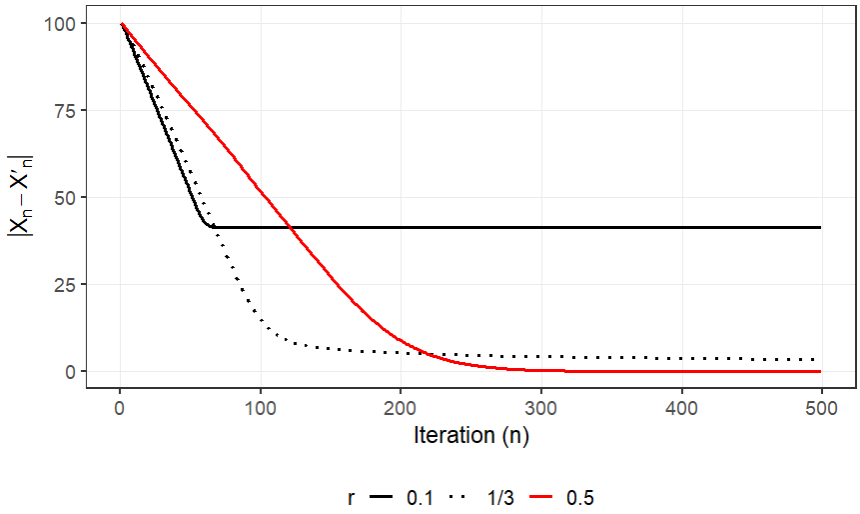


Figure 3: The simulated expected value of $|X_n - X'_n|$ where X_n, X'_n follow Zeno's walk and are simulated using CRN. The CRN estimate uses the mean of 1000 simulations. $X_0 = 0, X'_0 = 100$. Since CRN is an optimal coupling (Lemma 4.7), the simulations are unbiased estimates of the actual convergence rates.

4.3.3 Autoregressive normal process in $\mathbb{R}^q$

Example 4.3 (Autoregressive normal process in $\mathbb{R}^q$)**.** The random vector $\{\vec{X}_n\}_{n\geqslant 1} \in \mathbb{R}^q$ is an autoregressive normal process if for matrix A and random vector $\vec{\theta}_n \sim N(\vec{0}_q, \Sigma_q^2)$.

$$\vec{X}_n = A\vec{X}_{n-1} + \vec{\theta}_n$$

Theorem 4.13 of [35] shows, using a CRN joint coupling (see Section 6.1 for an explanation of the connection between contraction rates and CRN), that the AR process is geometrically ergodic when the maximum eigenvalue of A is less than 1. If A has only positive entries, then by Lemma 4.8 below, we show that CRN is an optimal coupling for the nth iteration of the Markov chain.

Lemma 4.8. *CRN is an optimal coupling for the first iteration of the autoregressive normal process in $\mathbb{R}^q$. If A has only positive entries, then CRN is an optimal coupling for the nth*

iteration of the autoregressive normal process in $\mathbb{R}^q$.

$$W_{||\cdot||_p,p}(\mathcal{L}(\vec{X}_n), \mathcal{L}(\vec{Y}_n)) = E[||f_n(\Theta_n, \vec{x}_0) - (f_n(\Theta_n, \vec{y}_0))||_p^p]^{1/p} \tag{8}$$

Further, if the maximum eigenvalue of A *is less than 1,* $||\vec{X}_n - \vec{Y}_n||_p \to 0$ *w.p. 1 when generated by CRN.*

Proof. CRN is the optimal coupling at the first iteration: For any fixed $i \in \{1, \ldots, q\}$ the coordinate $X_i = (A\vec{X}_0)_i + \theta_i$ is an increasing function of θ_i. So, by Theorem 4.4, CRN is an optimal coupling for the first iteration of the autoregressive normal process in $\mathbb{R}^q$.

CRN is the optimal coupling at the nth iteration: Fix $i, l \in \{1, \ldots q\}$ and $x_i < x_i'$. Since $a_{lj} \geqslant 0$ by assumption,

$$f^{(l)}(\vec{\theta}, (\vec{x}_{-i}, x_i)) = a_{li}x_i + \sum_{j \neq i} a_{lj}x_j + \theta_l \leqslant a_{li}x_i' + \sum_{j \neq i} a_{lj}x_j + \theta_l = f^{(l)}(\vec{\theta}, (\vec{x}_{-i}, x_i'))$$

Since and i, l were arbitrary, $A\vec{x} + \vec{\theta}$ is coordinate-wise monotone in $\vec{x}$. By Theorem 4.5 CRN is an optimal coupling on the nth iteration of the autoregressive normal process in $\mathbb{R}^q$ and equation 8 holds. Finally, Theorem 4.23 of [35] shows that if the maximum eigenvalue of A is less than 1, then the process is geometrically ergodic, so by Corollary 2, $||\vec{X}_n - \vec{Y}_n||_p^p \to 0$ w.p. 1 when generated by CRN. □

4.3.4 Bayesian location model Gibbs sampler

Example 4.4 (Bayesian location model Gibbs sampler)**.** Suppose that we are given data points $Y_1, \ldots, Y_J \sim N(\mu, \tau^{-1})$ where μ, τ^{-1} are unknown and $J \geqslant 3$. Let μ, τ^{-1} have flat priors on $\mathbb{R}$ and $\mathbb{R}_+$. The Gibbs algorithm is based on the marginal posterior distributions of μ, τ^{-1} and the conditional distributions are defined as follows.

- $\mu_{n+1} = \bar{y} + Z_{n+1}/\sqrt{J\tau_n}$
- $\tau_{n+1}^{-1} = \frac{\frac{S}{2} + \frac{J}{2}(\bar{y} - \mu_{n+1})^2}{G_{n+1}}$

Where $Z_n \sim N(0,1)$ and $G_n \sim \Gamma(\frac{J+2}{2}, 1)$ are independent and $S = \sum_{i=1}^{J}(y_i - \bar{y})^2$. $\Gamma(\alpha, \beta)$ represents the gamma distribution, $\Gamma^{-1}(\alpha, \beta)$ represents the inverse gamma distribution where α and β represent the shape and scale, respectively.

The variable τ^{-1} can be written as a random-coefficient auto-regressive process $\tau_n^{-1} = H_n G_n \tau_{n-1}^{-1} + G_n$ where $H_n \sim \Gamma\left(\frac{1}{2}, \frac{S}{2}\right)$ and $G_n \sim \Gamma^{-1}\left(\frac{J+2}{2}, \frac{S}{2}\right)$.

Theorem 6 of [30] and Theorem 4.14 of [35] show, via a CRN coupling (see Section 6.1 for the relationship between contraction rates and CRN), that the Gibbs sampler for the Bayesian location model has contraction rate of $\frac{1}{J}$.

Lemma 4.9. *CRN is an optimal coupling on the nth iteration of the Bayesian location model Gibbs sampler when k is QAC. That is for* $\vec{\theta}_n = (H_n, G_n)$,

$$W_k(\mathcal{L}(\tau_n^{-1}), \mathcal{L}(\tau_n^{-1'})) = E[k(f_n(\Theta_n, \tau_0^{-1}), f_n(\Theta_n, \tau_0^{-1'}))] \tag{9}$$

Further, $k(\tau_n^{-1}, \tau_n^{-1'}) \to 0$ *w.p. 1 when generated by CRN.*

Proof. Note first that $H_n, G_n, \tau_{n-1}^{-1} \geqslant 0$. For any $H_n = h, \tau_{n-1}^{-1} = \tau^{-1}$, the function $\tau_n^{-1} = hG_n\tau^{-1} + G_n$ is increasing with respect to G_n. For any $G_1 = g, \tau_{n-1}^{-1} = \tau^{-1}$, the function $\tau_1^{-1} = H_n g\tau^{-1} + g$ is increasing with respect to H_n. For any $H_n = h, G_n = g$, the function $\tau_1^{-1} = hg\tau_{n-1}^{-1} + g$ is increasing with respect to τ_{n-1}^{-1}. Thus, f is coordinate-wise monotone in terms of (H_n, G_n) and τ_n^{-1}.

So by Theorem 4.5, CRN is an optimal coupling on the nth iteration of the Bayesian location model Gibbs sampler when k a QAC norm and equation 9 follows.

Finally, Theorem 4.14 of [35] shows that the process is geometrically ergodic. By Corollary 2, $k(\tau_n^{-1}, \tau_n^{-1'}) \to 0$ w.p. 1 when generated by CRN. □

4.3.5 Linear ARCH process

Example 4.5 (Linear ARCH process)**.** Let $\{X_n\}_{n\geqslant 1} \in \mathbb{R}$ be a linear ARCH process. Then for i.i.d. θ_n

$$X_n = (\beta_0 + \beta_1 X_{n-1})\theta_n$$

We show that for the linear ARCH process, if $X_n \geqslant 0$ for all $n \in \mathbb{N}$, CRN is the joint coupling that generates the Wasserstein distance for iteration $n \geqslant 1$ and that if the process is geometrically ergodic, then the ℓ^p distance converges to 0 w.p. 1.

Lemma 4.10. *Suppose that $\beta_0, \beta_1 > 0$ and $\theta_n > 0$ a.s., then CRN is an optimal coupling on the nth iteration of the linear ARCH process when k is QAC.*

$$W_k(\mathcal{L}(X_n), \mathcal{L}(Y_n)) = E[k(f_n(\Theta_n, x_0), f_n(\Theta_n, y_0))] \tag{10}$$

Further, if $\beta_1 E[|\theta_0|] < 1$, $k(X_n, Y_n) \to 0$ w.p. 1 when generated by CRN.

Proof. Since $(\beta_0 + \beta_1 X_{n-1}) > 0$ w.p. 1, $X_n = f(\theta_n, X_{n-1})$ is always an increasing function of θ_n. Since $\beta_1\theta_n > 0$ w.p. 1, $f(\theta_n, X_{n-1})$ is always an increasing function of X_{n-1}. So by Theorem 4.5, CRN is an optimal coupling on the nth iteration of the linear ARCH process when k is QAC. Finally if $\beta_1 E[|\theta_0|] < 1$, then the process is geometrically ergodic (Theorem 5.1 of [35]), so by Corollary 2, $k(X_n, Y_n) \to 0$ w.p. 1 when generated using CRN. □

5 When monotone conditions are not satisfied

Monotonicity is important when establishing the optimal coupling between two random variables, because it orders the function with regard to what is relevant: the distance between $f(\theta, x)$ and $f(\theta, y)$ as a function of U where $\theta = f_\theta^{-1}(U)$. However, sometimes f is not monotone. In such cases we ask ourselves, is CRN the optimal coupling? Probably not. So, how could we generate a coupling that has lower expected distance? If we can propose an

equimeasurable rearrangement of f w.r.t U, f^*, that is monotone, then CRN applied to the proposed f^* would generate the optimal coupling.

In the following section, we propose two equimeasurable transformations $g : [0,1] \to [0,1]$ and $h : [0,1] \times \{0,1\} \to [0,1]$ that, when used correctly, generate monotone functions and consequently lower expected distances when simulated using CRN.

5.1 Equimeasurable functions

'Equimeasurable' when used within the context of CRN was discussed as early as 1934 [17]. The equimeasurable functions, however, were limited to the quantile transformation of $U \sim \text{Unif}(0,1)$, which is described in Section 3. Equimeasurable rearrangements were also defined in Chapter 1 of [39], and [8], but was again only defined in terms of quantile functions.

While the quantile function is an optimal transformation, in the problem setup with IRFs we do not know the CDF of X, but only know that $X = f(\theta, x)$ and that $\theta = F_\theta^{-1}(U)$ for known F_θ. Thus we're interested in generating an equimeasurable monotone rearrangment of $f(F_\theta^{-1}(U), x)$ with respect to U by creating a mapping $g : [0,1] \to [0,1]$ such that $f(F_\theta^{-1}(g(U)), x)$ is a monotone function of U and $\mathcal{L}(f(F_\theta^{-1}(g(U)), x)) = \mathcal{L}(f(F_\theta^{-1}(U), x))$.

In this section we study equimeasurable functions as transformations on the random variable U that preserve the measure of U (see Definition 5.1). This is similar to the approach taken in [34, 33], which studied generating lower bounds on eigenvalues using rearrangements of the eigenfunctions.

Definition 5.1 (Equimeasurable function of $f(U)$)**.** Let $f : [0,1] \to \mathbb{R}$ and $U \sim \text{Unif}(0,1)$. f^* is an equimeasurable function of f if $\mathcal{L}(f(U)) = \mathcal{L}(f^*(U))$.

An equimeasurable function f^* is specific to the function f and the random variable U. In this paper we restrict our definition to $U \sim \text{Unif}(0,1)$.

The following are two function transformations that can be applied to U. We can then use

these transformations to generate monotone equimeasurable transformations of our random function.

If the values of a function are flipped over an interval $(a, b) \in [0, 1]$, Lemma 5.1 states that this is an equimeasurable function.

Lemma 5.2 states that if I partition an interval into two parts $(a, b) = (a, t) \cup (t, b)$ then I can replace the interval (a, b) with (a, t) with probability $\frac{t-a}{b-a}$ and replace it with (t, b) with probability $\frac{b-t}{b-a}$, hence 'merging' two intervals.

Lemma 5.1 (Interval flip)**.** *Let* $(a, b) \subset [0, 1]$*,* $U \sim \text{Unif}(0, 1)$*, and the function* g *to be*

$$g(U) = \begin{cases} U & \text{if } U \notin (a, b) \\ a + b - U & \text{if } U \in (a, b) \end{cases} \tag{11}$$

Then $\mathcal{L}(f(U)) = \mathcal{L}(f(g(U)))$*. The proof can be found in Section 7.8.*

Lemma 5.2 (Interval merge)**.** *Let* $(a, b) \subset [0, 1]$*,* $t \in (a, b)$*,* $U \sim \text{Unif}(0, 1)$ *and the functions* $h(U, \xi)$ *be as follows for independent random variables* $U \sim \text{Unif}(0, 1)$ *and* $\xi \sim \text{Bernoulli}(\frac{t-a}{b-a})$

$$h(U, \xi) = \begin{cases} \xi \left(\frac{t-a}{b-a}(U - a) + a\right) + (1 - \xi) \left(\frac{b-t}{b-a}(U - a) + t\right) & \text{if } U \in (a, b) \\ U & \text{if } U \notin (a, b) \end{cases} \tag{12}$$

Then $\mathcal{L}(f(U)) = \mathcal{L}(f(h(U, \xi)))$*. The proof can be found in Section 7.9.*

5.2 Example

Numerical Example 5.1 (Random Logistics Map)**.** We revisit Example 4.1, to prove by counterexample that CRN is not the optimal coupling for a two-step random logistic map, even though it is the optimal one step coupling. Suppose that the initial values of the random

logistic map are $X_0 = \frac{1}{2}$ and $Y_0 = \frac{1}{8}$, then the second iteration of the functions are

$$X_2 = 4B_2B_1(1 - B_1) \qquad\qquad Y_2 = \frac{7}{4}B_2'B_1'\left(1 - \frac{7}{16}B_1'\right) \tag{13}$$

Recall $B_1, B_2 \sim \beta(\alpha + 1/2, \alpha - 1/2)$. Denote F_β to be the CDF of $\beta(\alpha + 1/2, \alpha - 1/2)$ so that $F_\beta^{-1}(U) \sim \beta(\alpha+1/2, \alpha-1/2)$. We can then re-write X_2 as follows where $U_1 \sim \text{Unif}(0, 1)$ and U_1, B_2 are independent,

$$X_2(U_1, B_2) = 4B_2F_\beta^{-1}(U_1)(1 - F_\beta^{-1}(U_1))$$

Observe that for any B_2, X_2 is an increasing function of U_1 when $U_1 \in [0, F_\beta(0.5)]$ and a decreasing function of U_1 when $U_1 \in [F_\beta(0.5), 1]$. We will construct an equimeasurable transform of $X_2(U_1, B_2)$, $X_2''(U_1, B_2, \xi)$, that is monotone in U_1 for $\xi \in \{0, 1\}$. To do so, we first generate an equimeasurable function, $X_2'(U_1, B_2) = X(g(U_1), B_2)$, that is increasing over $U_1 \in (F_\beta(0.5), 1]$ where $g : [0, 1] \to [0, 1]$ is defined as follows,

$$g(U) = \begin{cases} U & \text{if } U \in [0, F_\beta(0.5)] \\ 1 + F_\beta(0.5) - U & \text{if } U \in (F_\beta(0.5), 1] \end{cases} \tag{14}$$

By Lemma 5.1, $\mathcal{L}(X_2') = \mathcal{L}(X_2)$.

The interval flip is easiest to visualize in Figure 4. In graph (A), when $U_1 < F_\beta(0.5) \approx 0.288$ ($B_1 < 0.5$), $X_2(U_1, B_2) = X_2'(U_1, B_2)$, however when $U_1 > F_\beta(0.5)$ ($B_1 > 0.5$), $X_2(U_1, B_2)$ is represented by the black line while the transform $X_2'(U_1, B_2)$ is represented by the red line. Note that the transform $X_2'(U_1, B_2)$ is increasing when $U_1 > F_\beta(0.5)$.

The transform $X_2'(U_1, B_2)$ is increasing over $U_1 \in [0, F_\beta(0.5))$ and $U_1 \in (F_\beta(0.5), 1]$. However, it is not monotonically increasing. To generate an equimeasurable map that is monotonically increasing with respect to U, we use Lemma 5.2 and introduce the auxiliary random variable ξ.

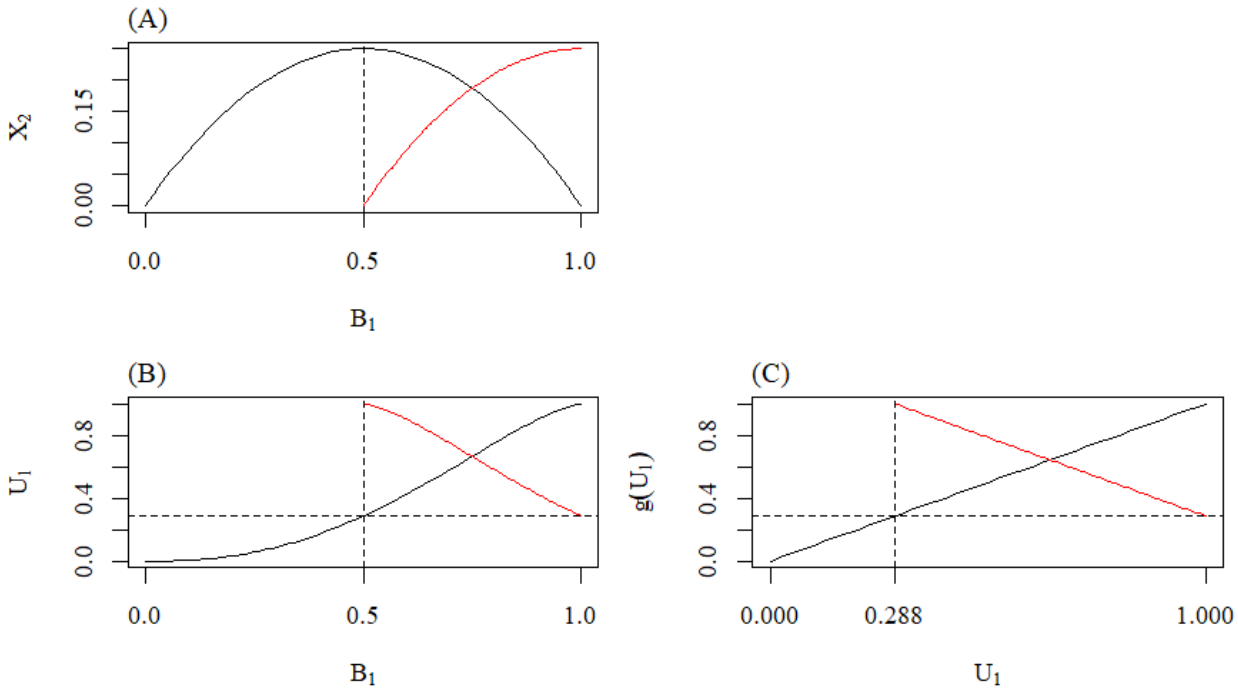


Figure 4: Set $\alpha = 2$ so that $B_1 \sim \beta(2.5, 1.5)$ and $F_\beta(0.5) = 0.288$. Set $B_2 = 1$ (its value is of little importance since it does not affect whether B_1 is increasing or not). In graph (A), the black line denotes $X_2(U_1, 1)$ and the red line denotes $X_2'(U_1, 1)$ when it is not equal to $X_2(U_1, 1)$. Graph (C) represents the transformation $g(U_1)$ in red compared to the identity function. Graph (B) then imposes the flip that was applied by g onto the values of B_1. Graph (A) shows the effect of the flip, g, through B_1 onto the value of X_2.

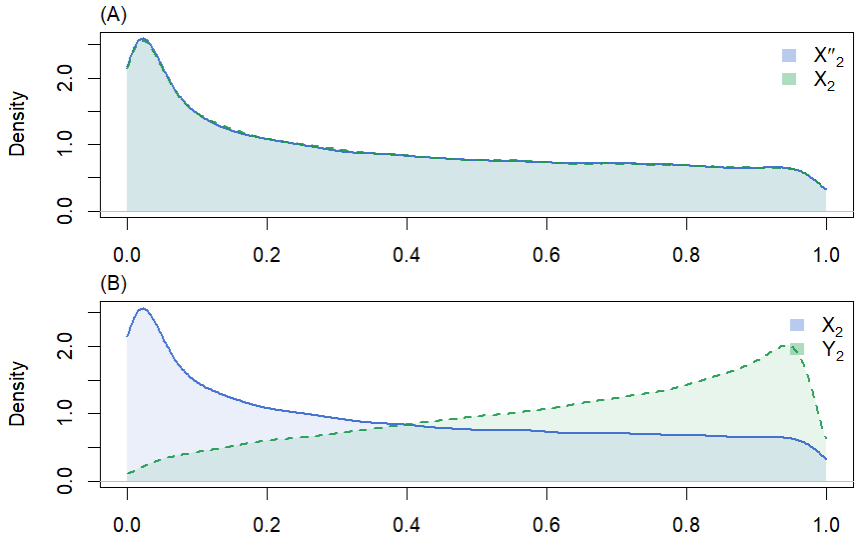


Figure 5: (A): Simulated density of $X_2 = 4B_2B_1(1-B_1)$ and $X_2''(U_1, B_2, \xi)$ (see Equation 15). The random variables $B_i \sim Beta(1.5, 0.5), i \in \{1, 2\}$, $U_1 \sim \text{Unif}(0, 1)$, $\xi \sim \text{Bernoulli}(F_\beta(0.5))$ are independent. This shows that the random variable X_2'', which is based on the transformations g and h are indeed equimeasurable to X_2. (B): Comparison of the density of X_2 and Y_2

Let $(a, b) = [0, 1]$ and $t = F_\beta(0.5)$. Then $X_2''(U_1, B_2, \xi) = X_2(g(h(U_1, \xi)), B_2)$ where $h(U_1, \xi)$ is generated according to Equation 12 is an equimeasurable function of U (Lemma 5.2). To summarize, X_2'' can be defined as follows for $\xi \sim \text{Bernoulli}(F_\beta(0.5))$

$$X_2''(U_1, B_2, \xi) = \begin{cases} 4B_2F_\beta^{-1}(1 - U_1(1 - F_\beta(0.5)))[1 - F_\beta^{-1}(1 - U_1(1 - F_\beta(0.5)))] & \text{if } \xi = 0 \\ 4B_2F_\beta^{-1}(F_\beta(0.5)U_1)[1 - F_\beta^{-1}(F_\beta(0.5)U_1)] & \text{if } \xi = 1 \end{cases} \tag{15}$$

Summarizing, Lemmas 5.2 and 5.1 imply that $X_2''(U_1, B_2, \xi)$ is an equimeasurable transform of $X_2(U_1, B_2)$. Figure 5 (A) graphs the simulated density of $X_2(U_1, B_2)$ and $X_2''(U_1, B_2, \xi)$, which visually shows that the densities are equal.

Finally, we want to show that CRN applied to $X_2(U_1, B_2)$ and $Y_2 = (U_1', B_2')$ is not the optimal coupling since X_2 is not an increasing function with respect to U_1. To do this we've simulated using CRN $X_2''(U_1, B_2, \xi)$, which is equimeasurable to $X_2(U_1, B_2)$, and $Y_2 = (U_1', B_2')$, which is increasing with respect to U_1' and B_2'. Figure 6 shows how the

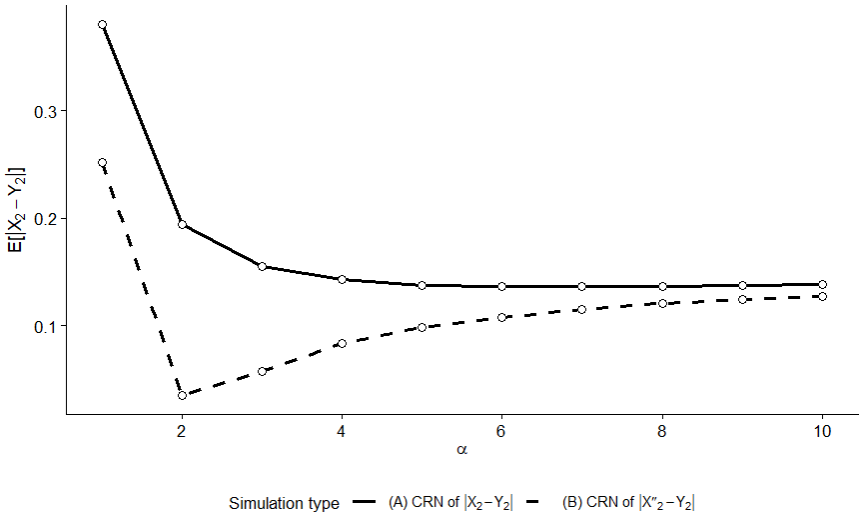


Figure 6: Line (A) represents the simulated expected distance of $|X_2(U_1, B_2) - Y_2(U_1, B_2)|$ ie. the expected distance when CRN is applied to the original function. Line (B) represents the simulated expected distance of $|X_2''(U_1, B_2, \xi) - Y_2(U_1, B_2)|$ ie. the expected distance when CRN is applied to the transformed $X_2''(U_1, B_2, \xi)$.

solid line, which is CRN applied to $|X_2(U_1, B_2) - Y_2(U_1, B_2)|$ has a larger expected distance compared to the transformed $|X_2''(U_1, B_2, \xi) - Y_2(F_\beta^{-1}(U_1), B_2)|$.

6 Connection between CRN and the backwards process

Backwards process The backwards process coupled with contraction conditions are commonly used to establish theorems that generate upper bounds on the Wasserstein distance between the distribution of a Markov chain and its corresponding stationary distribution (see [11, 38, 37]).

The backward process, $\{\tilde{X}_n\}_{n\geqslant 1}$, which is not a Markov chain [37] but tends to converge pointwise to a limit, is defined as follows,

$$\tilde{X}_n = f(\theta_1, (f(\theta_2, \dots f(\theta_n, X_0))) \tag{16}$$

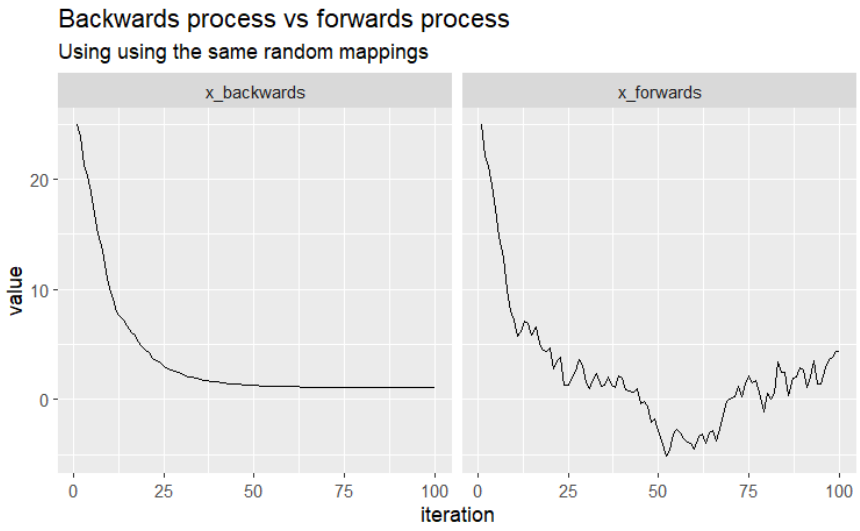


Figure 7: The backwards, $\{\tilde{X}_n\}_{n\geqslant 1}$, and forwards, $\{X_n\}_{n\geqslant 1}$, process of the autoregressive normal system where $X_n, \tilde{X}_n$ are simulated using the same random mappings, $X_n = 0.9X_{n-1} + Z_n, Z_n \sim N(0.1)$ and $X_0 = 25$.

The difference between the Markov chain and the backwards process is the order of $\theta_1, \theta_2, \ldots, \theta_n$ to generate the next iteration (Note the forward process, $X_n = f(\theta_n, (f(\theta_{n-1}, \ldots f(\theta_1, X_0)))$, compared to equation 16). Figure 7 graphs the forward and backward process of an autoregressive process using the same random mappings. The point towards which the backwards process converges is itself random and is distributed according to the stationary distribution.

Despite the difference in behaviour of the two processes, the marginal distributions of the forward and backward processes are the same, $X_n \stackrel{d}{=} \tilde{X}_n$. In fact, the following theorem shows that the expected distance over the joint distribution of $J(\tilde{X}_n, \tilde{X}_{n+1})$ realizes the same expected distance as the $J(X_n, Y_n)$ when CRN is used.

Theorem 6.1. *Let X_n, Y_n be two copies of a Markov chain such that $X_n = f(\theta_n, X_{n-1})$ and $Y_n = f(\theta'_n, Y_{n-1})$. Suppose that $X_0 = x$ and $Y_0 = f(\theta, x)$ where $\theta \perp\!\!\!\perp \theta_i$ for $i \in \{1 \ldots, n\}$, then the expected distance between X_n and Y_n using CRN ($\theta_i = \theta'_i$) is equal to the expected distance between consecutive iterations of the backwards process, $\bar{X}_n$ and $\bar{X}_{n+1}$.*

$$E[d(X_n, Y_n) \mid X_0 = x] = E[d(\tilde{X}_n, \tilde{X}_{n+1}) \mid \tilde{X}_0 = x]$$

Proof.

$$
\begin{aligned}
&E[d(X_n, Y_n) \mid X_0 = x] \\
&= E[d[f(\theta_n, f(\theta_{n-1}, \dots f(\theta_1, x)\dots)), f(\theta_n, f(\theta_{n-1}, \dots f(\theta_1, f(\theta, x))\dots))] \mid X_0 = x] \\
&= E[d[f(\theta_n, f(\theta_{n-1}, \dots f(\theta_1, x)\dots)), f(\theta_n, f(\theta_{n-1}, \dots f(\theta_1, f(\theta_{n+1}, x))\dots))] \mid X_0 = x] \\
&= E[d[f(\theta_1, f(\theta_2, \dots f(\theta_n, x)\dots)), f(\theta_1, f(\theta_2, \dots f(\theta_n, f(\theta_{n+1}, x))\dots))] \mid \tilde{X}_0 = x] \\
&= E[d(\tilde{X}_n, \tilde{X}_{n+1}) \mid \tilde{X}_0 = x]
\end{aligned}
$$

□

This means that for initial values defined as in Theorem 6.1, when CRN is an optimal coupling at iteration n, the expected distance between consecutive iterations of the backwards process is also the optimal coupling.

6.1 Connection between CRN and the contraction condition

Calculating contraction conditions are a common way to generate theoretical upper bounds on the Wasserstein distance between two copies of a Markov chain. The following is an example contraction condition. A survey of contraction conditions can be found in [38].

Definition 6.1 (Average contractive)**.** A Markov chain is average contractive if for all x, y $\pi-$everywhere,

$$\gamma_{cont} = \sup_{x \neq x'} \frac{E[k(f(\theta, x), f(\theta, x'))]}{k(x, x')} < 1$$

Notice how the contraction rate implicitly uses CRN. That is, we apply the same θ in $f(\theta, x)$ and $f(\theta, x')$ to calculate the contraction rate.

Implicitly using CRN to generate contraction conditions is not limited to average contraction rates. Anywhere where backwards processes are used to generate contraction rates, the CRN is implicitly assumed, by Theorem 6.1. Local contractivity (see Definition 5 of [37]) uses backwards processes and so by extension CRN. Theorem 5.1 of [11], which pro-

vides sufficient conditions for geometric convergence uses backwards processes in the proof and again by extension CRN. Backwards processes are used more often in proofs, because they facilitate convergence to stationarity.

7 Appendix

7.1 Proof of Proposition 3.3

Proof. Proof of Proposition 3.3

$\Longleftarrow$: $|F_n^{-1}(U) - F^{-1}(U)| \to 0$ w.p. 1 means that $F_n^{-1}(U) \to F^{-1}(U)$ almost surely, which means that $\mathcal{L}(F_n^{-1}(U)) \to \mathcal{L}(F^{-1}(U))$ in distribution. Since $F_n^{-1}(U) \sim \mathcal{L}(X_n)$ and $F^{-1}(U) \sim \mathcal{L}(X_\infty)$, this implies $\mathcal{L}(X_n) \to \mathcal{L}(X_\infty)$.

$\Longrightarrow$ This proof greatly resembles the proof of Theorem 10.1.1 (4) of [31].

Denote $D_{F^{-1}} \in [0,1]$ as the set of discontinuous points on F^{-1} and $C_{F^{-1}} \in [0,1]$ as the set of continuous points on F^{-1}.

Let $0 < u < 1$, $\epsilon > 0$, and choose ω such that $F^{-1}(u) - \epsilon < \omega < F^{-1}(u)$ and $\pi(\omega) = 0$. Then $F(\omega) < u$ and since $F_n(\omega) \to F(\omega)$, for sufficiently large n, $F_n(\omega) < u$ and so, $F^{-1}(u) - \epsilon < \omega < F_n^{-1}(u)$. Thus,

$$\liminf F_n^{-1}(u) \geqslant F^{-1}(u) \tag{17}$$

Let $0 < u < u' < 1$ and choose ω such that $F^{-1}(u') < \omega < F^{-1}(u') + \epsilon$ and $\pi(\omega) = 0$. Then $u < u' \leqslant F(F^{-1}(u')) \leqslant F(\omega)$ and since $F_n(\omega) \to F(\omega)$, for sufficiently large n, $u \leqslant F_n(\omega)$ and $F_n^{-1}(u) \leqslant \omega < F^{-1}(u') + \epsilon$. Thus,

$$\limsup F_n^{-1}(u) \leqslant F^{-1}(u') \tag{18}$$

By equation 17 and equation 18 if F^{-1} is continuous at u (ie. $u \in C_{F^{-1}}$), then $\lim_{n\to\infty} F_n^{-1}(u) = F^{-1}(u)$. Or in other words, $P(\lim_{n\to\infty} |F_n^{-1}(u) - F^{-1}(u)| = 0 \mid U \in C_{F^{-1}}) = 1$.

Next suppose that $U \sim Unif(0,1)$. Since F^{-1} is non-decreasing on $(0,1)$, it has countably many discontinuous points and so $P(U \in D_{F^{-1}}) = 0$ (and $P(U \in C_{F^{-1}}) = 1$).

Thus, the probability of $\lim_{n\to\infty} |F_n^{-1}(u) - F^{-1}(u)| = 0$ is as follows,

$$\begin{aligned}
&P(\lim_{n\to\infty} |F_n^{-1}(U) - F^{-1}(U)| = 0) \\
&\geqslant P(\lim_{n\to\infty} |F_n^{-1}(U) - F^{-1}(U)| = 0 \mid U \in C_{F^{-1}}) P(U \in C_{F^{-1}}) \\
&= 1
\end{aligned}$$

And so, when CRN is applied to the quantile function, $|F_n^{-1}(U) - F^{-1}(U)| \to 0$ w.p. 1. □

7.2 Proof of Proposition 4.1

Proof of Proposition 4.1. Denote $g(\theta) = f(\theta, x)$ and $h(\theta) = f(\theta, y)$ for fixed x and y.

Since g and h are monotone in θ by assumption, if $\theta_1 \leqslant \theta_1'$ and $\theta_2 \leqslant \theta_2'$, then $g(\theta_1) \leqslant g(\theta_1')$ and $h(\theta_2) \leqslant h(\theta_2')$. By the quasi-antitone property of k^p,

$$k\left(g(\theta_1), h(\theta_2)\right)^p + k\left(g(\theta_1'), h(\theta_2')\right)^p - k\left(g(\theta_1'), h(\theta_2)\right)^p - k\left(g(\theta_1), h(\theta_2')\right)^p \leqslant 0$$

And so, $k(g(\theta_1), h(\theta_2))^p$ is quasi-antitone with respect to θ_1 and θ_2.

Further, since g and h are right-continuous functions, $k(g(\theta_1), h(\theta_2))^p$ is right-continuous with respect to θ_1, θ_2 and so $k(g(\theta_1), h(\theta_2))^p$ is QAC with respect to θ_1, θ_2. We can then apply Proposition 3.1.

$$\begin{aligned}
W_p(\mathcal{L}(X), \mathcal{L}(Y)) &= \left(\inf_{\mathcal{J}(X,Y)} E\big[k(X,Y)^p\big] \right)^{1/p} \\
&= \left(\inf_{\mathcal{J}(\theta_1,\theta_2)} E\big[k(g(\theta_1), h(\theta_2))^p\big] \right)^{1/p} \\
&= \left(E\big[k(g(F_\theta^{-1}(U)), h(F_\theta^{-1}(U)))^p\big] \right)^{1/p} && \text{by Prop 3.1} \\
&= \left(E\big[k(g(\theta), h(\theta))^p\big] \right)^{1/p} \\
&= \left(E\big[k(f(\theta, x), f(\theta, y))^p\big] \right)^{1/p}
\end{aligned}$$

The second equality is true because $X = g(\theta)$ and $Y = h(\theta)$ by definition and so every coupling $J(\theta_1, \theta_2)$ maps to a coupling in $J(X, Y)$. □

7.3 Proof of Proposition 4.3

Proof. CRN is the coordinate-wise optimal coupling Since for all $i \in \{1, \ldots, d\}$ the function $f(\vec{\theta}, x)$ with fixed $\vec{\theta}_{-i}$ (μ_{-i}-everywhere) is an increasing function with respect to θ_i. And

since k is QAC, then by Proposition 4.1, for fixed θ_{-i}, x, y, CRN is an optimal coupling,

$$\inf_{\mathcal{J}(\mu_i,\mu_i)} E[k(f(\vec{\theta},x),f(\vec{\theta}',y)) \mid \vec{\theta}_{-i},\vec{\theta}'_{-i}] = E[k(f(\vec{\theta},x),f(\vec{\theta}',y)) \mid \vec{\theta}_{-i},\vec{\theta}'_{-i},\theta_i=\theta'_i] \tag{19}$$

We apply a proof by induction to show that CRN is an optimal coupling over the unconditional expected value. Let $\times_{n=1}^{d} J(\theta_n,\theta'_n) \in \mathcal{J}(\mu,\mu)$ be a joint distribution.

Base step:

The expected value of $k(f(\vec{\theta},x),f(\vec{\theta}',y))$ conditioned on coordinates $(\theta_{2:d},\theta'_{2:d}) \in \mathcal{J}(\mu_{2:d},\mu_{2:d})$ is bounded below by the expected value when CRN is applied to coordinate 1 ($\theta_1 = \theta'_1$). This is true by equation 19 for arbitrary joint distribution $J(\theta_1,\theta_1) \in \mathcal{J}(\mu_1,\mu_1)$.

$$E_{J(\theta_1,\theta_1)}[k(f(\vec{\theta},x),f(\vec{\theta}',y)) \mid \vec{\theta}_{2:d},\vec{\theta}'_{2:d}] \geqslant E_{\theta_1}[k(f(\vec{\theta},x),f(\vec{\theta}',y)) \mid \vec{\theta}_{2:d},\vec{\theta}'_{2:d},\theta_1=\theta'_1]$$

Induction step: Suppose that for $0 \leqslant N \leqslant d-1$, the joint expected distance conditioned on coordinates $(\vec{\theta}_{N+1:d},\vec{\theta}'_{N+1:d}) \in \mathcal{J}(\mu_{N+1:d},\mu_{N+1:d})$ is bounded below by the expected distance when CRN is applied to coordinates 1 to N ($\vec{\theta}_{1:N} = \vec{\theta}'_{1:N}$).

$$E_{\times_{n=1}^{N} J(\theta_n,\theta'_n)}[k(f(\vec{\theta},x),f(\vec{\theta}',y)) \mid \vec{\theta}_{N+1:d},\vec{\theta}'_{N+1:d}] \tag{20}$$

$$\geqslant E_{\theta_{1:N}}[k(f(\vec{\theta},x),f(\vec{\theta}',y)) \mid \vec{\theta}_{N+1:d},\vec{\theta}'_{N+1:d},\vec{\theta}_{1:N}=\vec{\theta}'_{1:N}] \tag{21}$$

We will then show that for $N+1$, the expected value of k conditioned on coordinates $(\vec{\theta}_{N+2:d},\vec{\theta}'_{N+2:d}) \in \mathcal{J}(\mu_{N+2:d},\mu_{N+2:d})$ is also attained when CRN is applied to coordinates 1

to $N+1$.

$$
\begin{aligned}
&E_{\times_{n=1}^{N+1} J(\theta_n,\theta'_n)}[k(f(\vec{\theta},x),f(\vec{\theta'},y)) \mid \vec{\theta}_{N+2:d},\vec{\theta'}_{N+2:d}] \\
&= E_{J(\theta_{N+1},\theta'_{N+1})}[E_{\times_{n=1}^{N} J(\theta_n,\theta'_n)}[k(f(\vec{\theta},x),f(\vec{\theta'},y)) \mid \vec{\theta}_{N+1:d},\vec{\theta'}_{N+1:d}] \mid \vec{\theta}_{N+2:d},\vec{\theta'}_{N+2:d}] \\
&\geqslant E_{J(\theta_{N+1},\theta'_{N+1})}[E_{\theta_{1:N}}[k(f(\vec{\theta},x),f(\vec{\theta'},y)) \mid \vec{\theta}_{N+1:d},\vec{\theta'}_{N+1:d},\vec{\theta}_{1:N}=\vec{\theta'}_{1:N}] \mid \vec{\theta}_{N+2:d},\vec{\theta'}_{N+2:d}] \quad \text{by eq. 20} \\
&= E_{J(\theta_{N+1},\theta'_{N+1})\times\vec{\theta}_{1:N}}[k(f(\vec{\theta},x),f(\vec{\theta'},y)) \mid \vec{\theta}_{N+2:d},\vec{\theta'}_{N+2:d},\vec{\theta}_{1:N}=\vec{\theta'}_{1:N}] \\
&\geqslant E_{\vec{\theta}_{1:N+1}}[k(f(\vec{\theta},x),f(\vec{\theta'},y)) \mid \vec{\theta}_{N+2:d},\vec{\theta'}_{N+2:d},\vec{\theta}_{1:N+1}=\vec{\theta'}_{1:N+1}] \quad \text{by eq. 19}
\end{aligned}
$$

The above implies that the expected value of k for joint distribution $\times_{n=1}^{d} J(\theta_n,\theta'_n) \in \mathcal{J}(\mu,\mu)$ is bounded below by the expected value of k when CRN is applied.

$$
\begin{aligned}
E_{\times_{n=1}^{d} J(\theta_n,\theta'_n)}[k(f(\vec{\theta},x),f(\vec{\theta'},y))] &\geqslant E_{\vec{\theta}_{1:d}}[k(f(\vec{\theta},x),f(\vec{\theta'},y)) \mid \vec{\theta}_{1:d}=\vec{\theta'}_{1:d}] &(22)\\
&= E_{\vec{\theta}}[k(f(\vec{\theta},x),f(\vec{\theta},y))] &(23)
\end{aligned}
$$

Since the joint distribution $\times_{n=1}^{d} J(\theta_n,\theta'_n) \in \mathcal{J}(\mu,\mu)$ was arbitrary,

$$\inf_{\mathcal{J}(\mu,\mu)} E[k(f(\vec{\theta},x),f(\vec{\theta'},y))] \geqslant E[k(f(\vec{\theta},x),f(\vec{\theta},y))]$$

Further, since $\times_{n=1}^{d} J(\theta_n,\theta_n) \in \mathcal{J}(\mu,\mu)$, the inequality goes the other way,

$$\inf_{\mathcal{J}(\mu,\mu)} E[k(f(\vec{\theta},x),f(\vec{\theta'},y))] \leqslant E[k(f(\vec{\theta},x),f(\vec{\theta},y))]$$

And so, equality ensues, $\inf_{\mathcal{J}(\mu,\mu)} E[k(f(\vec{\theta},x),f(\vec{\theta'},y))] = E[k(f(\vec{\theta},x),f(\vec{\theta},y))]$. Finally, since $\inf_{\mathcal{J}(\mu,\mu)} E[k(f(\vec{\theta},x),f(\vec{\theta'},y))] = W_p(\mathcal{L}(f(\theta,x)),\mathcal{L}(f(\theta,y))) = W_p(\mathcal{L}(X),\mathcal{L}(Y))$, we get equation 5.

$$W_p(\mathcal{L}(X),\mathcal{L}(Y)) = E[k(f(\vec{\theta},x),f(\vec{\theta},y))]$$

□

7.4 Proof of Theorem 4.4

Proof.

$$\inf_{\mathcal{J}(\mu,\mu)} E[||f(\vec{\theta},\vec{x}) - f(\vec{\theta'},\vec{y})||_p^p] = \inf_{\mathcal{J}(\mu,\mu)} \sum_{i=1}^{q} E[(f^{(i)}(\vec{\theta},\vec{x}) - f^{(i)}(\vec{\theta'},\vec{y}))^p] \tag{24}$$

$$\geqslant \sum_{i=1}^{q} \inf_{\mathcal{J}(\mu,\mu)} E[(f^{(i)}(\vec{\theta},\vec{x}) - f^{(i)}(\vec{\theta'},\vec{y}))^p] \tag{25}$$

$$= \sum_{i=1}^{q} E[(f^{(i)}(\vec{\theta},\vec{x}) - f^{(i)}(\vec{\theta},\vec{y}))^p] \qquad \text{by Equation 5} \tag{26}$$

Since $\inf_{\mathcal{J}(\mu,\mu)} E[||f(\vec{\theta},\vec{x}) - f(\vec{\theta'},\vec{y})||_p^p] \leqslant \sum_{i=1}^{q} E[(f^{(i)}(\vec{\theta},\vec{x}) - f^{(i)}(\vec{\theta},\vec{y}))^p]$, equality follows and we get Equation 6,

$$\begin{aligned} W_{||\cdot||_p,p}(\mathcal{L}(\vec{X}),\mathcal{L}(\vec{Y})) &= \left(\inf_{\mathcal{J}(\mu,\mu)} E[||f(\vec{\theta},\vec{x}) - f(\vec{\theta'},\vec{y})||_p^p]\right)^{1/p} \\ &= \left(\sum_{i=1}^{q} E[(f^{(i)}(\vec{\theta},\vec{x}) - f^{(i)}(\vec{\theta},\vec{y}))^p]\right)^{1/p} \qquad \text{By eq. 24} \\ &= \left(E[||f^{(i)}(\vec{\theta},\vec{x}) - f^{(i)}(\vec{\theta},\vec{y})||_p^p]\right)^{1/p} \end{aligned}$$

□

7.5 Proof of Theorem 4.5

Proof. Recall $\vec{\theta} \in \mathbb{R}^d$, $\vec{X} \in \mathbb{R}^q$

Fix $j \in \{1 \dots, d\}$. Define $\vec{\theta}_{1,-j} = \theta_1 \backslash \theta_{1,j}$.

Base step: If $\theta_{1,j} \leqslant \theta'_{1,j}$ μ_j-a.e., then for each $k \in \{1 \dots q\}$ and $\vec{\theta}_{1,-j} \sim \mu_{-j}$, $f^{(k)}((\vec{\theta}_{1,-j}, \theta_{1,j}), \vec{x}) \leqslant f^{(k)}((\vec{\theta}_{1,-j}, \theta'_{1,j}), \vec{x})$ by monotonicity of $\vec{\theta}$. Thus for arbitrary $l \in \{1 \dots q\}$,

$$
\begin{aligned}
&f^{(l)}(\theta_2, f((\vec{\theta}_{1,-j}, \theta_{1,j}), \vec{x})) \\
&= f^{(l)}[\theta_2, (f^{(1)}((\vec{\theta}_{1,-j}, \theta_{1,j}), \vec{x}), f^{(2)}((\vec{\theta}_{1,-j}, \theta_{1,j}), \vec{x}), \dots f^{(q)}((\vec{\theta}_{1,-j}, \theta_{1,j}), \vec{x}))] \\
&\leqslant f^{(l)}[\theta_2, (f^{(1)}((\vec{\theta}_{1,-j}, \theta'_{i,j}), \vec{x}), f^{(2)}((\vec{\theta}_{1,-j}, \theta_{1,j}), \vec{x}), \dots f^{(q)}((\vec{\theta}_{1,-j}, \theta_{1,j}), \vec{x}))] \quad \text{Since } f^{(l)} \text{ is monotone in } x_1 \\
&\leqslant f^{(l)}[\theta_2, (f^{(1)}((\vec{\theta}_{1,-j}, \theta'_{i,j}), \vec{x}), f^{(2)}((\vec{\theta}_{1,-j}, \theta'_{1,j}), \vec{x}), \dots f^{(q)}((\vec{\theta}_{1,-j}, \theta_{1,j}), \vec{x}))] \quad \text{Since } f^{(l)} \text{ is monotone in } x_2 \\
&\dots \\
&\leqslant f^{(l)}[\theta_2, (f^{(1)}((\vec{\theta}_{1,-j}, \theta'_{1,j}), \vec{x}), f^{(2)}((\vec{\theta}_{1,-j}, \theta'_{1,j}), \vec{x}), \dots f^{(q)}((\vec{\theta}_{1,-j}, \theta'_{1,j}), \vec{x}))] \quad \text{Since } f^{(l)} \text{ is monotone in } x_q \\
&= f^{(l)}[\theta_2, f((\vec{\theta}_{1,-j}, \theta'_{1,j}), \vec{x})]
\end{aligned}
$$

Since j was arbitrary, $f(\vec{\theta_2}, f(\vec{\theta_1}, \vec{x}))$, is monotone in $\vec{\theta_1}$. In other words, $f_2(\Theta_2, \vec{x})$ is monotone in Θ_2 where $\Theta_2 = (\vec{\theta_1}, \vec{\theta_2})$.

Induction step: Next, suppose that $f_n(\Theta_n, \vec{x})$ is monotone in Θ_n. Show that $f_{n+1}(\Theta_{n+1}, \vec{x})$ is monotone in Θ_{n+1}

Fix $i \in \{1 \dots, n\}$ and $j \in \{1 \dots, d\}$. Define θ_j as the jth coordinate of the Θ_n. Define $\Theta_{n,-j} = \Theta_n \backslash \theta_j$. If $\theta_j \leqslant \theta'_j$ μ_i-a.e., then for each $l \in \{1 \dots q\}$, $\Theta_{n,-j} \sim \mu_{-j} \times \mu^{n-1}$, $\vec{x}$ π-a.e.

The following inequality holds, because f_n is monotone in Θ_n and f is monotone in θ.

$$
\begin{aligned}
f^{(l)}_{n+1}((\Theta_{n+1,-j}, \theta_j), \vec{x})) &= f^{(l)}(\vec{\theta}_{n+1}, f_n((\Theta_{n,-j}, \theta_j), \vec{x})) \\
&\leqslant f^{(l)}(\vec{\theta}_{n+1}, f_n((\Theta_{n,-j}, \theta'_j), \vec{x})) \\
&= f^{(l)}_{n+1}(\Theta_{n+1}, \vec{x})
\end{aligned}
$$

Since j was arbitrary, $f_{n+1}(\Theta_{n+1}, \vec{x})$, is monotone in Θ_{n+1} if $f_n(\Theta_n, \vec{x})$, is monotone in Θ_n.

By induction and base step, $f_n(\Theta_n, \vec{x})$ is monotone in Θ_n, $n \geqslant 1$. Thus we apply Proposition 4.3 if $x \in \mathbb{R}$ or Theorem 4.4 if $\vec{x} \in \mathbb{R}^q$ to obtain Equation 7. □

7.6 Proof of Corollary 1

Proof. Since $\vec{X}_n, \vec{Y}_n$ converge to π in $P_k(\mathbb{R}^q)$, $W_p(\mathcal{L}(\vec{X}_n), \pi) \to 0$ and $W_p(\pi, \mathcal{L}(\vec{Y}_n)) \to 0$ by Proposition 3.2. Next, since $W_p(\mathcal{L}(\vec{X}_n), \mathcal{L}(\vec{Y}_n)) \leqslant W_p(\mathcal{L}(\vec{X}_n), \pi) + W_p(\pi, \mathcal{L}(\vec{Y}_n))$,

$$W_p(\mathcal{L}(\vec{X}_n), \mathcal{L}(\vec{Y}_n)) \to 0$$

Further, since the conditions of Theorem 4.5 are satisfied for any $n \geqslant 1$, CRN is an optimal coupling, $W_p(\mathcal{L}(\vec{X}_n), \mathcal{L}(\vec{Y}_n)) = E[k(f_n(\Theta_n, \vec{x}), f_n(\Theta_n, \vec{y}))]$ and so,

$$E[k(f_n(\Theta_n, \vec{x}), f_n(\Theta_n, \vec{y}))] \to 0$$

Fix $\epsilon > 0$. By Markov's inequality,

$$\lim_{n\to\infty} P(k(f_n(\Theta_n, \vec{x}), f_n(\Theta_n, \vec{y})) \geqslant \epsilon) \leqslant \lim_{n\to\infty} \frac{E[k(f_n(\Theta_n, \vec{x}), f_n(\Theta_n, \vec{y}))]}{\epsilon} = \frac{0}{\epsilon} = 0$$

Thus, $\{\vec{X}_n\}_{n\geqslant 1}, \{\vec{Y}_n\}_{n\geqslant 1}$ generated using CRN will converge in probability. □

7.7 Proof of Corollary 2

Proof. If a Markov chain is geometrically ergodic, then there exists a function $M : \mathbb{R}^q \to \mathbb{R}$ and a $\gamma < 1$ such that for any $\vec{x} \in \mathbb{R}^q$, $W_p(\mathcal{L}(\vec{X}_n \mid \vec{X}_0 = \vec{x}_0), \pi) \leqslant M(\vec{x}_0)\gamma^n$. This means that

for starting values $\vec{x}_0, \vec{y}_0$,

$$\begin{aligned} W_p(\mathcal{L}(\vec{X}_n \mid \vec{X}_0 = \vec{x}_0), \mathcal{L}(\vec{Y}_n \mid \vec{Y}_0 = \vec{y}_0)) &\leqslant W_p(\mathcal{L}(\vec{X}_n \mid \vec{X}_0 = \vec{x}_0), \pi) + W_p(\pi, \mathcal{L}(\vec{Y}_n \mid \vec{Y}_0 = \vec{y}_0)) \\ &\leqslant M(\vec{x}_0)\gamma^n + M(\vec{y}_0)\gamma^n \\ &\leqslant 2\max\{M(\vec{x}_0), M(\vec{y}_0)\}\gamma^n \end{aligned}$$

Since the conditions of Theorem 4.5 are satisfied, $E[k(f_n(\Theta_n, \vec{x}_0), f_n(\Theta_n, \vec{y}_0))] = W_p(\mathcal{L}(\vec{X}_n \mid \vec{X}_0 = \vec{x}_0), \mathcal{L}(\vec{Y}_n \mid \vec{Y}_0 = \vec{y}_0))$ and so,

$$E[k(f_n(\Theta_n, \vec{x}_0), f_n(\Theta_n, \vec{y}_0))] \leqslant 2\max\{M(\vec{x}_0), M(\vec{y}_0)\}\gamma^n$$

Let $\epsilon > 0$

$$\begin{aligned} \sum_{n=0}^{\infty} P(k(f_n(\Theta_n, \vec{x}_0), f_n(\Theta_n, \vec{y}_0)) > \epsilon) &\leqslant \sum_{n=0}^{\infty} \frac{E[k(f_n(\Theta_n, \vec{x}_0), f_n(\Theta_n, \vec{y}_0))]}{\epsilon} \\ &\leqslant \sum_{n=0}^{\infty} \frac{2\max\{M(\vec{x}_0), M(\vec{y}_0)\}\gamma^n}{\epsilon} \\ &= \frac{2\max\{M(\vec{x}_0), M(\vec{y}_0)\}}{\epsilon(1-\gamma)} \\ &< \infty \end{aligned}$$

By the Borel-Cantelli Lemma (Theorem 3.4.2 of [31]), $P(k(f_n(\Theta_n, \vec{x}_0), f_n(\Theta_n, \vec{y}_0)) > \epsilon \text{ i.o.}) = 0$ for $\epsilon > 0$, which is equivalent to saying that $P(\lim_{n\to\infty} k(f_n(\Theta_n, \vec{x}_0), f_n(\Theta_n, \vec{y}_0)) = 0) = 1$ by Lemma 5.2.1 of [31] and so $\vec{X}_n, \vec{Y}_n$ generated using CRN will converge almost surely. □

7.8 Proof of Lemma 5.1

Proof of Lemma 5.1. Let $t \in \mathbb{R}$. We want to show that $P(f(g(U)) \leqslant t) = P(f(U) \leqslant t)$

$$
\begin{aligned}
P(f(g(U)) \leqslant t) &= \int_{[0,1]} I(f(g(u)) \leqslant t)\mathrm{d}u \\
&= \int_a^b I(f(a+b-u) \leqslant t)\mathrm{d}u + \int_{(a,b)^C} I(f(u) \leqslant t)\mathrm{d}u \\
&= -\int_b^a I(f(v) \leqslant t)\mathrm{d}v + \int_{(a,b)^C} I(f(u) \leqslant t)\mathrm{d}u \\
&= \int_a^b I(f(u) \leqslant t)\mathrm{d}u + \int_{(a,b)^C} I(f(u) \leqslant t)\mathrm{d}u \\
&= \int_{[0,1]} I(f(u) \leqslant t)\mathrm{d}u \\
&= P(f(U) \leqslant t)
\end{aligned}
$$

□

7.9 Proof of Lemma 5.2

Proof of Lemma 5.2. Let $x \in \mathbb{R}$. We want to show that $P(f(h(U,\xi)) \leqslant x) = P(f(U) \leqslant x)$.

First note the following integrals.

$$
\begin{aligned}
&\int_{(a,b)} I(f(h(u,1)) \leqslant x)\mathrm{d}u && (27)\\
&= \int_{(a,b)} I\left(f\left(\frac{u-a}{b-a}(t-a)+a\right) \leqslant x\right)\mathrm{d}u && (28)\\
&= \int_{(a,t)} I\left(f\left(v\right) \leqslant x\right)\frac{b-a}{t-a}\mathrm{d}v \qquad \text{by a change of variable } v = \frac{u-a}{b-a}(t-a)+a && (29)
\end{aligned}
$$

$$\int_{(a,b)} I(f(h(u,0)) \leqslant x)\mathrm{d}u \tag{30}$$

$$= \int_{(a,b)} I\left(f\left(\frac{u-a}{b-a}(b-t)+t\right) \leqslant x\right)\mathrm{d}u \tag{31}$$

$$= \int_{(t,b)} I\left(f\left(v\right) \leqslant x\right)\frac{b-a}{b-t}\mathrm{d}v \qquad \text{by a change of variable } v = \frac{u-a}{b-a}(b-t)+t \tag{32}$$

We show that $P(f(h(U,\xi)) \leqslant x) = P(f(U) \leqslant x)$ as follows,

$$\begin{aligned}
&P(f(h(U,\xi)) \leqslant x) \\
&= P(f(h(U,\xi)) \leqslant x \mid \xi = 1)P(\xi = 1) + P(f(g(U,\xi)) \leqslant x \mid \xi = 0)P(\xi = 0) \\
&= P(f(h(U,1)) \leqslant x)\frac{t-a}{b-a} + P(f(h(U,0)) \leqslant x)\frac{b-t}{b-a} \\
&= \left(\int_{(a,b)} I(f(h(u,1)) \leqslant x)\mathrm{d}u + \int_{(a,b)^c} I(f(h(u,1)) \leqslant x)\mathrm{d}u\right)\frac{t-a}{b-a} + \\
&\qquad \left(\int_{(a,b)} I(f(h(u,0)) \leqslant x)\mathrm{d}u + \int_{(a,b)^c} I(f(h(u,0)) \leqslant x)\mathrm{d}u\right)\frac{b-t}{b-a} \\
&= \int_{(a,b)} I(f(h(u,1)) \leqslant x)\mathrm{d}u\frac{t-a}{b-a} + \int_{(a,b)} I(f(h(u,0)) \leqslant x)\mathrm{d}u\frac{b-t}{b-a} + \\
&\qquad \int_{(a,b)^c} I(f(u) \leqslant x)\mathrm{d}u \\
&= \int_{(a,t)} I\left(f\left(v\right) \leqslant x\right)\frac{b-a}{t-a}\mathrm{d}v\frac{t-a}{b-a} + \int_{(t,b)} I\left(f\left(v\right) \leqslant x\right)\frac{b-a}{b-t}\mathrm{d}v\frac{b-t}{b-a} + \\
&\qquad \int_{(a,b)^c} I(f(u) \leqslant x)\mathrm{d}u \qquad \text{By eq. 27, 30} \\
&= \int_{(a,t)} I\left(f\left(v\right) \leqslant x\right)\mathrm{d}v + \int_{(t,b)} I\left(f\left(v\right) \leqslant x\right)\mathrm{d}v + \int_{(a,b)^c} I(f(u) \leqslant x)\mathrm{d}u \\
&= \int_{[0,1]} I(f(u) \leqslant x)\mathrm{d}u \\
&= P(f(U) \leqslant x)
\end{aligned}$$

□

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